

**Frank, 71** · 50 pack-years · "barrel chest" · pursed-lip breathing · prolonged expiration with wheezes · O<sub>2</sub> sat 88% on room air · FEV<sub>1</sub>/FVC 0.55 · **COPD exacerbation**

BIO 004 · WEEK 3 · CASE DEEP DIVE

# The Respiratory System

A retired mechanic, 50 years of smoke, lungs that can't exhale fast

## PATIENT CASE

**Frank is 71**, retired auto mechanic, smoked a pack a day from age 16 to 65 (about **50 pack-years**). He has known COPD. He comes in today with **three days of worsening cough, increased green sputum, and more shortness of breath**, especially when walking from the parking lot.

On exam he is **tripoding, pursed-lip breathing, with a barrel-shaped chest**. Auscultation reveals **diminished breath sounds bilaterally with prolonged expiration and diffuse wheezes**. O<sub>2</sub> sat is **88% on room air**. CXR shows hyperinflated lungs with flattened diaphragms. PFTs from last month: **FEV<sub>1</sub>/FVC 0.55**, total lung capacity increased, residual volume increased. He is admitted for a **COPD exacerbation**.

COPD is what happens when decades of smoke destroy the alveolar walls (emphysema) and inflame the airways (chronic bronchitis). Understanding why Frank can't exhale and why his lungs are getting bigger requires knowing exactly how the respiratory system is built, from the nose to the alveoli, and what each part does.

## GUIDING QUESTION

What is the path of air from nose to alveoli, what makes the conducting airways different from the respiratory airways, and why does destruction of alveolar walls trap air inside lungs that should empty easily?

## PREWORK

Companion to your Week 3 Respiratory System prework page. Terms, video, and spaced recall for upper and lower airway anatomy, larynx, lungs and pleurae, gross structure, alveolar structure, and ventilation mechanics live there.

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## UPPER RESPIRATORY

# Nose to larynx

The respiratory system is divided into **upper** (nose, nasal cavity, pharynx) and **lower** (larynx, trachea, bronchi, lungs). Air goes through all of them on every breath.

### Nose & nasal cavity

Warms, humidifies, and filters air. Lined by pseudostratified ciliated columnar epithelium (the mucociliary escalator). Conchae increase surface area.

### Paranasal sinuses

Frontal, ethmoid, sphenoid, maxillary. Lighten the skull, warm and humidify air, drain into the nasal cavity.

### Pharynx

Common passage for air and food. Three regions: nasopharynx (air only), oropharynx (both), laryngopharynx (both, splits below).

### Larynx

"Voice box." Cartilage framework (thyroid, cricoid, epiglottis, paired arytenoids) housing the vocal cords. Guards the airway from food entry (epiglottis closes during swallowing) and produces sound.

# Trachea down to alveoli

From the larynx, air enters the **trachea** (10-12 cm long, supported by C-shaped cartilage rings open posteriorly toward the esophagus). The trachea bifurcates at the **carina** (~T4/T5, the sternal angle level) into right and left main bronchi.

- **Right main bronchus:** wider, shorter, more vertical than left. Aspirated objects favor the right side.
- **Left main bronchus:** narrower, longer, more horizontal. Passes under the aortic arch.

From there, the airway branches repeatedly (~23 generations of branching):

## • Main bronchi → lobar bronchi → segmental bronchi

Still cartilage-supported. Air conducting only — no gas exchange.

## • Bronchioles

< 1 mm diameter. No cartilage. Walls are smooth muscle (which is why they bronchoconstrict in asthma).

## • Terminal bronchioles

Last of the purely conducting airways.

## • Respiratory bronchioles

First place gas exchange happens (a few alveoli bud off).

## • Alveolar ducts → alveolar sacs → alveoli

The terminal gas-exchange units. ~300 million alveoli per adult.

## Where actual gas exchange happens

An **alveolus** is a thin-walled air sac, about 250-300  $\mu\text{m}$  in diameter, wrapped in a dense capillary network. Three cell types in the alveolar wall:

### ● Type I pneumocytes

Squamous cells that form ~95% of the alveolar surface. So thin that  $\text{O}_2$  and  $\text{CO}_2$  diffuse across them in milliseconds.

### ● Type II pneumocytes

Cuboidal cells that secrete *surfactant*, a phospholipid mixture that reduces surface tension and prevents alveolar collapse on exhalation. Premature infants are surfactant-deficient → respiratory distress syndrome.

### ● Alveolar macrophages

Phagocytose particles and bacteria that reach the alveoli. The lung's last line of defense after the mucociliary escalator.

The **respiratory membrane** (where  $\text{O}_2$  and  $\text{CO}_2$  actually cross) is just three layers: alveolar epithelium (Type I cell), fused basement membranes, capillary endothelium. Total thickness about 0.5  $\mu\text{m}$ . Diseases that thicken this membrane (fibrosis, pulmonary edema) impair gas exchange.

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## Gross structure and the membranes that wrap them

The right lung has **three lobes** (upper, middle, lower) separated by horizontal and oblique fissures. The left lung has **two lobes** (upper, lower) separated by an oblique fissure — the heart takes the space where the middle lobe would be (instead, there's a notch and a tongue-shaped projection of the upper lobe called the *lingula*).

Each lung is enveloped by two **pleural membranes**:



### Visceral pleura

Covers the lung surface.



### Parietal pleura

Lines the thoracic wall, diaphragm, and mediastinum.

The thin space between them is the **pleural cavity**, containing a thin film of serous fluid. The negative intrapleural pressure keeps the lung inflated against the chest wall. If air enters the pleural space (*pneumothorax*), the lung collapses.

## How a breath actually happens

**Inspiration** is active. The *diaphragm* contracts and flattens, the *external intercostals* lift the ribs up and out, and accessory muscles (sternocleidomastoid, scalenes) help in distress. The thoracic cavity expands. Intrapleural pressure becomes more negative. The lungs are pulled outward, alveolar pressure drops below atmospheric, and air flows in.

**Expiration** at rest is passive. The diaphragm and external intercostals relax. The elastic recoil of the lung tissue (and the chest wall in some conditions) pulls everything back. Alveolar pressure rises above atmospheric, air flows out.

Forced expiration uses abdominal muscles and internal intercostals. This becomes critical in obstructive disease like Frank's.

## Why FEV1/FVC tells you so much

Lung disease falls into two big patterns:

- **Obstructive:** airflow OUT is impaired. FEV1 (volume exhaled in 1 second) drops more than FVC (total volume exhaled). **FEV1/FVC ratio < 0.70**. Examples: asthma, COPD, chronic bronchitis, emphysema. *Frank's pattern*.
- **Restrictive:** lung CAN'T EXPAND fully. Both FEV1 and FVC drop proportionally. FEV1/FVC ratio is preserved or increased. Examples: pulmonary fibrosis, kyphoscoliosis, neuromuscular weakness.

**What COPD does anatomically.** Two processes, often together. *Emphysema*: alveolar walls are destroyed; small airways lose the tethering that normally holds them open during exhalation; they collapse on expiration, trapping air. The lung becomes hyperinflated. *Chronic bronchitis*: chronic inflammation thickens bronchial walls and produces excess mucus, narrowing the airway and providing a substrate for infection. Both produce obstruction.



**DRAW ZONE 1** -- Diagram the path of air from nose to alveolus. Label upper airway structures, trachea, carina, main/lobar/segmental bronchi, bronchioles, alveoli.

Mark where cartilage stops (bronchioles) and where gas exchange starts (respiratory bronchioles).



**DRAW ZONE 2** -- Sketch a normal alveolus and a destroyed (emphysematous) alveolus. Show the loss of alveolar walls, reduced surface area, and the collapsing small airway during exhalation.

Show why a barrel chest develops: chronic air trapping flattens the diaphragm and pushes the chest into a permanent inspiratory shape.



## RETURNING TO FRANK

# Why he can't exhale, why his chest looks the way it does, why pursed lips help

Frank's exam findings are a directory of COPD pathophysiology. Each one maps to specific damage in his lungs.

### **"Barrel-shaped chest"**

Chronic air trapping holds the lungs in a persistently inflated state. The diaphragm flattens, the ribs sit elevated, the AP diameter of the chest increases. The chest is stuck in inspiration.

### **"Pursed-lip breathing"**

Maintains positive pressure in the airways during exhalation. This positive pressure splints open the small airways (which would otherwise collapse from loss of alveolar tethering) and prolongs expiration.

### **"Prolonged expiration with wheezes"**

Narrowed airways (mucus, smooth muscle constriction, loss of tethering) generate turbulent flow during the slow exhalation. Wheezes are the audible sign.

### **"Diminished breath sounds"**

Hyperinflated lungs are full of trapped air that doesn't move with each breath. Air movement is reduced; breath sounds are quieter.

### **"FEV1/FVC 0.55"**

Diagnostic of airflow obstruction. He exhales only 55% of his vital capacity in the first second; a normal person exhales > 70%. Below 0.70 = obstructive disease.

### **"O2 sat 88%"**

Loss of alveolar surface area (emphysema) + ventilation-perfusion mismatch (some alveoli are damaged, others get less air). Plus the current exacerbation adds inflammation, mucus, possibly infection on top of his baseline disease.

**Airway, alveolus, surface area, recoil.** Frank's lungs are showing you the price of fifty years of smoke at every level of the respiratory system. When you go back to the prework, the alveolar diagram will land differently — it's exactly what he's missing.